

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Vectors in Two and Three Dimensions

Week 4 Day 2 AM



COURSE SURVEY

The SSILP Mathematics Survey is now open until 10 Aug. 2026 (11.59pm), you should have received an email on this. Please follow the link provided in the email to complete the survey. Your feedback is very important for us to improve the course.



OUTLINE

① Three-Dimensional Coordinate Geometry

② Vectors in Three Dimensions

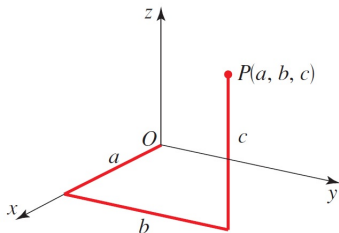
③ The Cross Product

④ Equations of Lines and Planes

⑤ Summary

THE 3D RECTANGULAR COORDINATE SYSTEM

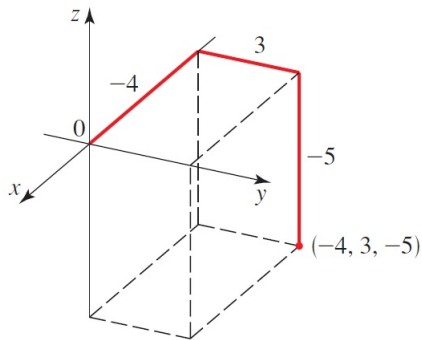
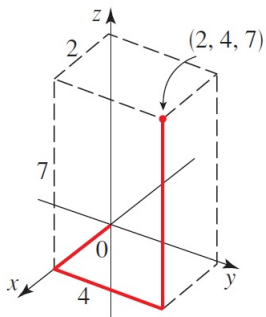
To represent points in 3D space, we extend the Cartesian plane by adding the z -axis that is perpendicular to the Cartesian plane. By convention, we put the x - and y -axes horizontally and the z -axis vertically, and we draw the orientation of the axes by the right-hand rule (right fingers curled from the positive x -axis to the positive y -axis and the right thumb pointing in the positive z -axis) as follows:



Now any point P in space can be located by a unique ordered triple of real numbers (a, b, c) , as shown above. a, b, c are the x, y, z -coordinates of P .

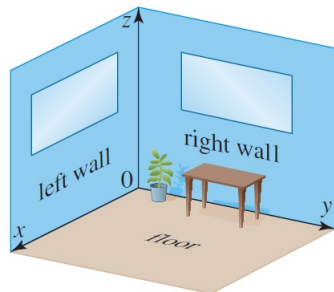
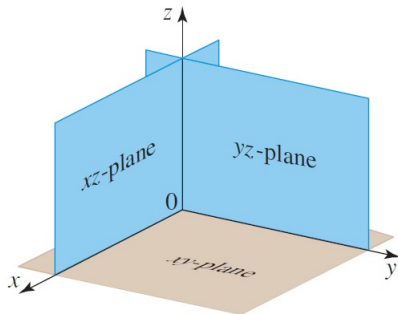
THE 3D RECTANGULAR COORDINATE SYSTEM - EXAMPLE

E.g.: The figures below show the position of points $(2, 4, 7)$ and $(-4, 3, -5)$.



THE 3D RECTANGULAR COORDINATE SYSTEM

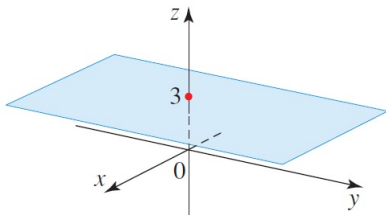
The ***xy-plane*** is the plane that contains the x - and y -axes; the ***yz-plane*** is the plane that contains the y - and z -axes; the ***xz-plane*** is the plane that contains the x - and z -axes. These three coordinate planes divide space into eight parts, called ***octants***. There is a **convention** of numbering the octants as octant I, II, ..., VIII, according to the signs of the three coordinates.



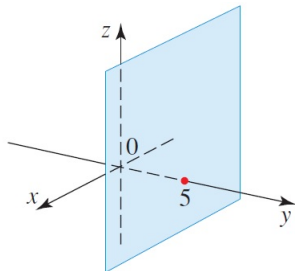
SURFACES IN 3D SPACE

In two-dimensional geometry the graph of an equation involving x and y is a curve in the plane. In three-dimensional geometry an equation in x , y , and z represents a surface in space. For instance, the xy -plane is given by the equation $z = 0$; the yz -plane is given by the equation $x = 0$; the xz -plane is given by the equation $y = 0$.

E.g.: Here are two more examples.



The plane $z = 3$

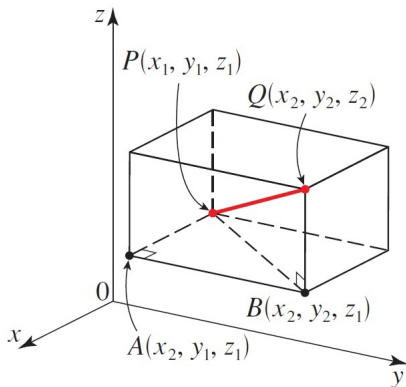


The plane $y = 5$

DISTANCE FORMULA IN THREE DIMENSIONS

By applying Pythagoras' theorem twice, we can compute the distance between the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in the space with the following formula:

$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

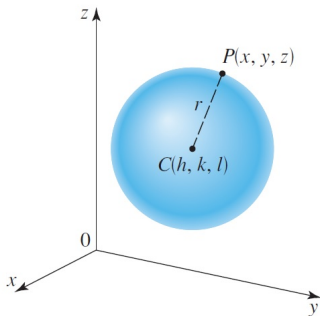


THE EQUATION OF A SPHERE IN THREE DIMENSIONS

EQUATION OF A SPHERE

An equation of a sphere with center $C(h, k, l)$ and radius r is

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$



The equation comes from $\sqrt{(x - h)^2 + (y - k)^2 + (z - l)^2} = r$, which literally says (x, y, z) are points at distance $= r$ from the point (h, k, l) .

THE EQUATION OF A SPHERE IN THREE DIMENSIONS

The general form of a sphere is the expanded version of the standard form from the previous slide:

$$x^2 + ax + y^2 + by + z^2 + cz + d = 0.$$

In order to find the center and the radius of the sphere, simply complete the square w.r.t. x , y and z separately to convert the general form back to the standard form.

E.g.: Given $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$, we first group the x, y, z terms respectively, after which we complete the square:

$$x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$$

$$\Leftrightarrow (x^2 + 4x) + (y^2 - 6y) + (z^2 + 2z) = -6$$

$$\Leftrightarrow (x^2 + 4x + 4) + (y^2 - 6y + 9) + (z^2 + 2z + 1) = -6 + 4 + 9 + 1$$

$$\Leftrightarrow (x + 2)^2 + (y - 3)^2 + (z + 1)^2 = 8$$

$$\Leftrightarrow (x - (-2))^2 + (y - 3)^2 + (z - (-1))^2 = (\sqrt{8})^2$$

The sphere is centered at $(-2, 3, -1)$ with radius $= \sqrt{8}$.

ACTIVITY 1

Show that the equations below represent spheres, and find their centers and radii.

(a) $x^2 + y^2 + z^2 - 10x + 2y + 8z = 9.$

(b) $\frac{1}{2}x^2 + \frac{1}{2}y^2 + \frac{1}{2}z^2 = 6x + y.$

ACTIVITY 1: SOLUTION

$$\begin{aligned} \text{(a)} \quad & x^2 + y^2 + z^2 - 10x + 2y + 8z = 9 \\ \Leftrightarrow & x^2 - 10x + 25 + y^2 + 2y + 1 + z^2 + 8z + 16 = 9 + 25 + 1 + 16 \\ \Leftrightarrow & (x - 5)^2 + (y + 1)^2 + (z + 4)^2 = 51 \\ \Leftrightarrow & (x - 5)^2 + (y - (-1))^2 + (z - (-4))^2 = (\sqrt{51})^2. \end{aligned}$$

Thus, it's a sphere centered at $(5, -1, -4)$ with radius $\sqrt{51}$.

$$\begin{aligned} \text{(b)} \quad & \frac{1}{2}x^2 + \frac{1}{2}y^2 + \frac{1}{2}z^2 = 6x + y \\ \Leftrightarrow & x^2 + y^2 + z^2 = 12x + 2y \\ \Leftrightarrow & x^2 - 12x + 36 + y^2 - 2y + 1 + z^2 = 36 + 1 \\ \Leftrightarrow & (x - 6)^2 + (y - 1)^2 + z^2 = 37 \\ \Leftrightarrow & (x - 6)^2 + (y - 1)^2 + (z - 0)^2 = (\sqrt{37})^2. \end{aligned}$$

Thus, it's a sphere centered at $(6, 1, 0)$ with radius $\sqrt{37}$. Restricted

OUTLINE

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- ② Vectors in Three Dimensions
- ③ The Cross Product
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VECTORS IN THREE DIMENSIONS

DEFINITION:

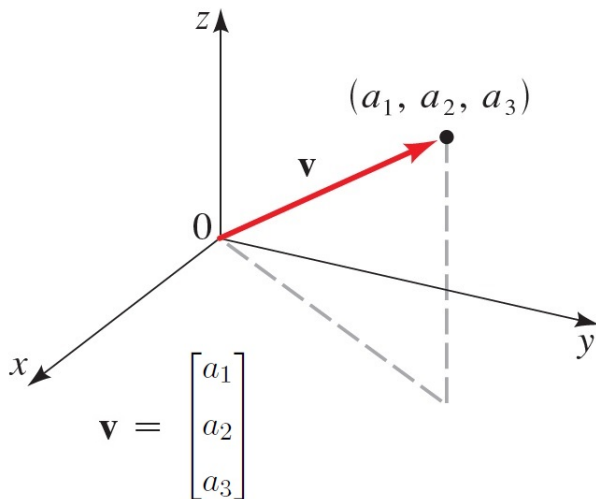
If a vector $\vec{v} = \overrightarrow{PQ}$ is represented in the 3D space with initial point $P(x_1, y_1, z_1)$ and terminal point (arrowhead) $Q(x_2, y_2, z_2)$, then we represent $\vec{v}$ as the column ordered triple

$$\vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{bmatrix},$$

where $x = x_2 - x_1$ is the ***x-component*** of $\vec{v}$, $y = y_2 - y_1$ is the ***y-component*** of $\vec{v}$, and $z = z_2 - z_1$ is the ***z-component*** of $\vec{v}$.

Alternative notations: We also represent $\vec{v}$ as $\langle x, y, z \rangle$ or $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

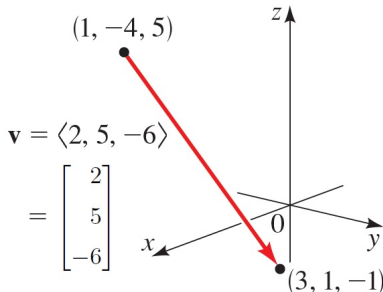
VECTORS IN THREE DIMENSIONS



VECTORS IN THREE DIMENSIONS- EXAMPLE

E.g.: The vector $\vec{v}$ has an initial point $P(1, -4, 5)$ and terminal point $Q(3, 1, -1)$, then

$$\vec{v} = \begin{bmatrix} 3 - 1 \\ 1 - (-4) \\ -1 - 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ -6 \end{bmatrix}.$$



MAGNITUDE/LENGTH

DEFINITION:

The *magnitude/length* the vector $\vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is

$$||\vec{v}|| = \sqrt{x^2 + y^2 + z^2}.$$

This is just a consequence of the distance formula in 3D.

E.g.: The vector $\vec{u} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$ has length

$$||\vec{u}|| = \sqrt{3^2 + 2^2 + 5^2} = \sqrt{38}.$$

ALGEBRAIC OPERATIONS ON VECTORS IN 3D

Vector addition, subtraction and scalar multiplication are defined for vectors in 3D in a similar fashion as the vectors in 2D.

DEFINITION:

For any $\vec{u} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ and $c \in \mathbb{R}$, we define the vector addition, subtraction and scalar multiplication as follows:

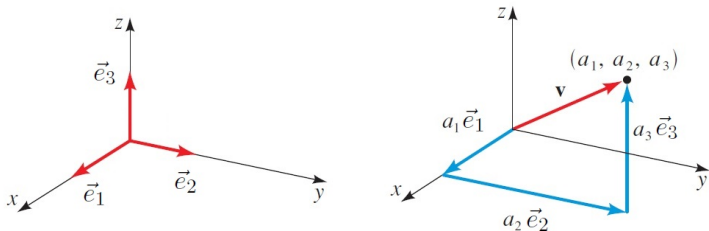
$$\vec{u} + \vec{v} := \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{bmatrix}, \quad \vec{u} - \vec{v} := \begin{bmatrix} a_1 - b_1 \\ a_2 - b_2 \\ a_3 - b_3 \end{bmatrix}, \quad c\vec{u} = \begin{bmatrix} ca_1 \\ ca_2 \\ ca_3 \end{bmatrix}.$$

The geometrical interpretations and properties for these operations are the same as the case of 2D vectors.

UNIT VECTOR

Similar to vectors in 2D, for vectors in 3D, we define the following three useful unit vectors:

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$



Any vector in 3D can be expressed in terms of $\vec{e}_1, \vec{e}_2, \vec{e}_3$ by

$$\vec{v} = a_1\vec{e}_1 + a_2\vec{e}_2 + a_3\vec{e}_3.$$

Remark: In physics, $\vec{e}_1, \vec{e}_2, \vec{e}_3$ are denoted by $\hat{i}, \hat{j}, \hat{k}$ respectively.

THE DOT PRODUCT FOR VECTORS IN 3D

We define the dot product for vectors in 3D in a similar fashion as vectors in 2D. All the properties of the dot product, including the dot product theorem for vectors in 2D, hold for vectors in 3D.

DEFINITION:

If $\vec{u} = a_1\vec{e}_1 + a_2\vec{e}_2 + a_3\vec{e}_3$ and $\vec{v} = b_1\vec{e}_1 + b_2\vec{e}_2 + b_3\vec{e}_3$, then their **dot product**, denoted by $\vec{u} \cdot \vec{v}$, is the **real number** defined by

$$\vec{u} \cdot \vec{v} := a_1b_1 + a_2b_2 + a_3b_3.$$

E.g.:

$$(2\vec{e}_1 - 3\vec{e}_2 - 1\vec{e}_3) \cdot (-\vec{e}_1 + 2\vec{e}_2 + 8\vec{e}_3) = (2)(-1) + (-3)(2) + (-1)(8) = -16.$$

E.g.:

$$(-\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (6\hat{i} + 5\hat{j} - 1\hat{k}) = (-1)(6) + (2)(5) + (3)(-1) = 1.$$

THE DOT PRODUCT THEOREM AND ORTHOGONAL VECTORS IN 3D

THE DOT PRODUCT THEOREM

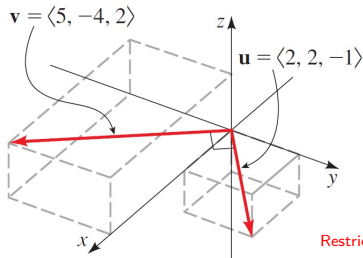
If θ is the angle between two non-zero vectors $\vec{u}$ and $\vec{v}$ in 3D, then

$$\vec{u} \cdot \vec{v} = ||\vec{u}|| ||\vec{v}|| \cos \theta.$$

DEFINITION:

Two vectors $\vec{u}$ and $\vec{v}$ in 3D are **orthogonal/perpendicular** iff.
 $\vec{u} \cdot \vec{v} = 0$.

E.g.: If $\vec{u} = 2\vec{e}_1 + 2\vec{e}_2 - \vec{e}_3$,
 $\vec{v} = 5\vec{e}_1 - 4\vec{e}_2 + 2\vec{e}_3$, then $\vec{u} \cdot \vec{v} =$
 $(5)(2) + (-4)(2) + (2)(-1) = 0$,
so the two vectors are orthogonal.



THE COMPONENT AND PROJECTION OF VECTORS IN 3D

Again, the component of $\vec{u}$ along $\vec{v}$ and the projection of $\vec{u}$ on $\vec{v}$ are defined similarly in 3D, with the exact same geometrical meanings.

DEFINITION:

For $\vec{u}, \vec{v}$ in 3D, the **component of $\vec{u}$ along/in the direction of $\vec{v}$** is the **real number**

$$\text{comp}_{\vec{v}}(\vec{u}) := \|\vec{u}\| \cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|},$$

where θ is the angle between the two vectors.

DEFINITION:

For $\vec{u}, \vec{v}$ in 3D, the **projection of $\vec{u}$ onto $\vec{v}$** is the **vector**

$$\text{proj}_{\vec{v}}(\vec{u}) := \text{comp}_{\vec{v}}(\vec{u}) \left(\frac{1}{\|\vec{v}\|} \vec{v} \right) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v}.$$

VECTORS IN HIGHER DIMENSIONS

In linear algebra, we will generalize vectors to n dimensions for any natural number n . All the vector related concepts such as components, magnitude/length, vector addition, subtraction and scalar multiplication, the properties of the operations on vectors, unit vectors $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n$, dot product and its properties, angle between vectors, the dot product theorem, the component of $\vec{u}$ along $\vec{v}$ and the projection of $\vec{u}$ on $\vec{v}$, etc. are all defined or behave similarly just like how we extend them from 2D to 3D.

ACTIVITY 2 [DO IT AT HOME]

- (a) Find the angle between $\vec{u}$ and $\vec{v}$.
- (i) $\vec{u} = 2\vec{e}_1 + 2\vec{e}_2 - \vec{e}_3$, $\vec{v} = \vec{e}_1 + 2\vec{e}_2 + 2\vec{e}_3$.
 - (ii) $\vec{u} = 3\hat{i} - 2\hat{j} + 4\hat{k}$, $\vec{v} = \hat{i} + \hat{j} + \hat{k}$.
- (b) Determine whether or not the given vectors are orthogonal.
- (i) $\vec{u} = 4\vec{e}_1 - 2\vec{e}_2 + 4\vec{e}_3$, $\vec{v} = \vec{e}_1 - 2\vec{e}_2 + 2\vec{e}_3$.
 - (ii) $\vec{u} = 4\hat{j} - \hat{k}$, $\vec{v} = 9\hat{i} + 2\hat{j} + 8\hat{k}$.
- (c) Given $\vec{u} = 3\vec{e}_1 - 2\vec{e}_2 + 4\vec{e}_3$. Express $\vec{u} = \vec{x} + \vec{y}$, where $\vec{x}$ is parallel to $\vec{v} = \vec{e}_1 + \vec{e}_2 + \vec{e}_3$ and $\vec{y}$ is orthogonal to $\vec{v}$.

ACTIVITY 2: SOLUTION

(a) (i)

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{||\vec{u}|| ||\vec{v}||} = \frac{(2)(1) + (2)(2) + (-1)(2)}{\sqrt{2^2 + 2^2 + (-1)^2} \sqrt{1^2 + 2^2 + 2^2}}$$
$$\Rightarrow \theta = \arccos\left(\frac{4}{9}\right) \approx 1.1102 \text{ rad} \approx 63.61^\circ.$$

(ii)

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{||\vec{u}|| ||\vec{v}||} = \frac{(3)(1) + (-2)(1) + (4)(1)}{\sqrt{3^2 + (-2)^2 + 4^2} \sqrt{1^2 + 1^2 + 1^2}}$$
$$\Rightarrow \theta = \arccos\left(\frac{5}{\sqrt{87}}\right) \approx 1.005 \text{ rad} \approx 57.58^\circ.$$

(b) (i) $\vec{u} \cdot \vec{v} = (4)(1) + (-2)(-2) + (4)(2) = 16 \neq 0$, so they are not orthogonal.

(ii) $\vec{u} \cdot \vec{v} = (0)(9) + (4)(2) + (-1)(8) = 0$, so they are orthogonal.

ACTIVITY 2: SOLUTION

(c)

$$\begin{aligned}\vec{x} &= \text{proj}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} \\ &= \frac{(3)(1) + (-2)(1) + (4)(1)}{(1^2 + 1^2 + 1^2)} (\vec{e}_1 + \vec{e}_2 + \vec{e}_3) \\ &= \frac{5}{3} (\vec{e}_1 + \vec{e}_2 + \vec{e}_3).\end{aligned}$$

$$\begin{aligned}\vec{y} &= \vec{u} - \vec{x} \\ &= 3\vec{e}_1 - 2\vec{e}_2 + 4\vec{e}_3 - \frac{5}{3} (\vec{e}_1 + \vec{e}_2 + \vec{e}_3) \\ &= \frac{4}{3}\vec{e}_1 - \frac{11}{3}\vec{e}_2 + \frac{7}{3}\vec{e}_3.\end{aligned}$$

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THE CROSS PRODUCT FOR VECTORS IN 3D

Although vectors in 3D inherit most properties and structures from vectors in 2D, there's one new important feature that emerges from vectors in 3D, which is the cross product.

In the 3D space, it's often useful to find a non-zero vector that is orthogonal to two given non-parallel vectors (this problem doesn't make sense in 2D). The cross product of the two non-parallel vectors will give us such a vector.

DEFINITION:

If $\vec{u} = a_1\vec{e}_1 + a_2\vec{e}_2 + a_3\vec{e}_3$ and $\vec{v} = b_1\vec{e}_1 + b_2\vec{e}_2 + b_3\vec{e}_3$ are three-dimensional vectors, then their **cross product/vector product**, denoted by $\vec{u} \times \vec{v}$, is the **vector** defined by

$$\vec{u} \times \vec{v} := (a_2b_3 - a_3b_2)\vec{e}_1 + (a_3b_1 - b_3a_1)\vec{e}_2 + (a_1b_2 - a_2b_1)\vec{e}_3.$$

THE CROSS PRODUCT FOR VECTORS IN 3D

Here's a mnemonic of the formula of the cross product:

- To compute the x -component $+a_2b_3 - a_3b_2$ of the cross product, **cover all the x -components**, “cross multiply and subtract” the remaining components:

$$\begin{array}{cc} \cancel{a_1} & \times & \cancel{b_1} \\ a_2 & & b_2 \\ a_3 & & b_3 \end{array}$$

$-a_3b_2$ $+a_2b_3$

- To compute the z -component $+a_1b_2 - a_2b_1$ of the cross product, **cover all the z -components**, “cross multiply and subtract” the remaining components:

$$\begin{array}{cc} \cancel{a_1} & \times & \cancel{b_1} \\ a_2 & & b_2 \\ \cancel{a_3} & & \cancel{b_3} \end{array}$$

$-a_2b_1$ $+a_1b_2$

THE CROSS PRODUCT FOR VECTORS IN 3D

- Lastly, to compute the y -component $-(+a_1b_3 - a_3b_1)$ of the cross product, **cover all the y -components**, “cross multiply and subtract” the remaining components and **multiply the result by -1** :

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$-a_3b_1$ $+a_1b_3$

Fun fact: The notion of cross product that takes in two vectors and returns a vector that is orthogonal to the two vectors is only defined for vectors in 3D and 7D.

PROPERTIES OF THE CROSS PRODUCT

By using the definition of the cross product, we can easily prove the following properties.

PROPERTIES OF THE CROSS PRODUCT

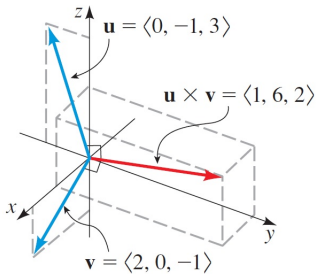
- $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$.
- $(c\vec{u}) \times \vec{v} = c(\vec{u} \times \vec{v}) = \vec{u} \times (c\vec{v})$.
- $(\vec{u} + \vec{v}) \times \vec{w} = \vec{u} \times \vec{w} + \vec{v} \times \vec{w}$.
- $\vec{u} \times \vec{u} = \vec{0}$.
- $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$.

Remark: Algebraically, the last property is equivalent to $(\vec{u} \times \vec{v}) \cdot \vec{u} = (\vec{u} \times \vec{v}) \cdot \vec{v} = 0$.

PROPERTIES OF THE CROSS PRODUCT - EXAMPLE

E.g.: If $\vec{u} = -\vec{e}_2 + 3\vec{e}_3$ and $\vec{v} = 2\vec{e}_1 - \vec{e}_3$, then a vector that is orthogonal to $\vec{u}$ and $\vec{v}$ is

$$\begin{aligned}\vec{u} \times \vec{v} &= \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix} \times \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} (-1)(-1) - (3)(0) \\ (3)(2) - (0)(-1) \\ (0)(0) - (-1)(2) \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix}.\end{aligned}$$



Note that there are only two unit vectors that are orthogonal to $\vec{u}$ and $\vec{v}$, which are

$$\frac{1}{\sqrt{1^2 + 6^2 + 2^2}} \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{41}} \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix} \quad \text{and} \quad -\frac{1}{\sqrt{41}} \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix}.$$

ACTIVITY 3

(a) For the given vectors $\vec{u}$ and $\vec{v}$, find a unit vector orthogonal to them.

(i) $\vec{u} = \vec{e}_1 + \vec{e}_2 - \vec{e}_3$, $\vec{v} = -\vec{e}_1 + \vec{e}_2 - \vec{e}_3$.

(ii) $\vec{u} = 3\hat{j} + 5\hat{k}$, $\vec{v} = -\hat{i} + 2\hat{k}$.

(b) Find a non-zero vector that is orthogonal to the plane containing the three given points.

(i) $P(6, -2, 1)$, $Q(5, -3, -1)$, $R(7, 0, 0)$.

(ii) $P(3, 4, 5)$, $Q(1, 2, 3)$, $R(1, -1, 1)$.

ACTIVITY 3: SOLUTION

(a) (i)

$$\begin{aligned}\vec{u} \times \vec{v} &= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \times \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} (1)(-1) - (-1)(1) \\ (-1)(-1) - (1)(-1) \\ (1)(1) - (1)(-1) \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}.\end{aligned}$$

A unit vector orthogonal to $\vec{u}$ and $\vec{v}$ is

$$\frac{1}{\|\vec{u} \times \vec{v}\|}(\vec{u} \times \vec{v}) = \frac{1}{\sqrt{0^2 + 2^2 + 2^2}} \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

ACTIVITY 3: SOLUTION

(a) (ii)

$$\begin{aligned}\vec{u} \times \vec{v} &= \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix} \times \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} (3)(2) - (5)(0) \\ (5)(-1) - (0)(2) \\ (0)(0) - (3)(-1) \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \\ 3 \end{bmatrix}.\end{aligned}$$

A unit vector orthogonal to $\vec{u}$ and $\vec{v}$ is

$$\frac{1}{\|\vec{u} \times \vec{v}\|}(\vec{u} \times \vec{v}) = \frac{1}{\sqrt{6^2 + (-5)^2 + 3^2}} \begin{bmatrix} 6 \\ -5 \\ 3 \end{bmatrix} = \frac{1}{\sqrt{70}} \begin{bmatrix} 6 \\ -5 \\ 3 \end{bmatrix}.$$

ACTIVITY 3: SOLUTION

(b) (i) Such a vector is

$$\begin{aligned}\overrightarrow{PQ} \times \overrightarrow{PR} &= \begin{bmatrix} 5 - 6 \\ -3 - (-2) \\ -1 - 1 \end{bmatrix} \times \begin{bmatrix} 7 - 6 \\ 0 - (-2) \\ 0 - 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -2 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} (-1)(-1) - (-2)(2) \\ (-2)(1) - (-1)(-1) \\ (-1)(2) - (-1)(1) \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \\ -1 \end{bmatrix}.\end{aligned}$$

(ii) Such a vector is

$$\begin{aligned}\overrightarrow{PQ} \times \overrightarrow{PR} &= \begin{bmatrix} 1 - 3 \\ 2 - 4 \\ 3 - 5 \end{bmatrix} \times \begin{bmatrix} 1 - 3 \\ -1 - 4 \\ 1 - 5 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix} \times \begin{bmatrix} -2 \\ -5 \\ -4 \end{bmatrix} \\ &= \begin{bmatrix} (-2)(-4) - (-2)(-5) \\ (-2)(-2) - (-2)(-4) \\ (-2)(-5) - (-2)(-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ 6 \end{bmatrix}.\end{aligned}$$

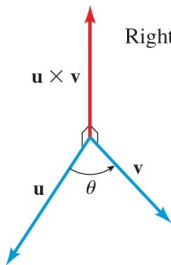
GEOMETRY OF THE CROSS PRODUCT

In fact, we can be more precise and explicit in describing $\vec{u} \times \vec{v}$:

THE CROSS PRODUCT THEOREM

Given $\vec{u}$ and $\vec{v}$ in 3D with angle between them being θ , $\vec{u} \times \vec{v}$ is the vector of length $||\vec{u}|| ||\vec{v}|| \sin \theta$ and orthogonal to $\vec{u}$ and $\vec{v}$ with the orientation given by the right-hand rule: right fingers curled from the direction of $\vec{u}$ to the direction of $\vec{v}$ and the right thumb pointing in the direction of $\vec{u} \times \vec{v}$. In particular, $\vec{u}$, $\vec{v}$ are **parallel** iff. $\vec{u} \times \vec{v} = \vec{0}$.

Please refer to page 663 of the textbook for the proof.



Right-hand rule

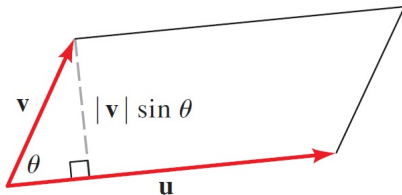


$$||\mathbf{u} \times \mathbf{v}|| = ||\mathbf{u}|| ||\mathbf{v}|| \sin \theta$$

APPLICATION - AREA OF A PARALLELOGRAM

Since the area of a parallelogram determined by $\vec{u}$ and $\vec{v}$ is

$$A = \text{base} \times \text{height} = \|\vec{u}\| \times \|\vec{v}\| \sin \theta = \|\vec{u} \times \vec{v}\|,$$



we have the following application of the cross product.

AREA OF A PARALLELOGRAM

The area of a parallelogram determined by $\vec{u}$ and $\vec{v}$ is $\|\vec{u} \times \vec{v}\|$.

AREA OF A PARALLELOGRAM - EXAMPLE

E.g.: The area of the triangle with vertices $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$ is half the area of the parallelogram determined by $\overrightarrow{PQ}$ and $\overrightarrow{PR}$, so

$$\begin{aligned}\text{area of the triangle} &= \frac{1}{2} \left\| \overrightarrow{PQ} \times \overrightarrow{PR} \right\| \\ &= \frac{1}{2} \left\| \begin{bmatrix} -2 - 1 \\ 5 - 4 \\ -1 - 6 \end{bmatrix} \times \begin{bmatrix} 1 - 1 \\ -1 - 4 \\ 1 - 6 \end{bmatrix} \right\| = \frac{1}{2} \left\| \begin{bmatrix} -3 \\ 1 \\ -7 \end{bmatrix} \times \begin{bmatrix} 0 \\ -5 \\ -5 \end{bmatrix} \right\| \\ &= \frac{1}{2} \left\| \begin{bmatrix} (1)(-5) - (-7)(-5) \\ (-7)(0) - (-3)(-5) \\ (-3)(-5) - (1)(0) \end{bmatrix} \right\| = \frac{1}{2} \left\| \begin{bmatrix} -40 \\ -15 \\ 15 \end{bmatrix} \right\| \\ &= \frac{1}{2} \sqrt{(40)^2 + (-15)^2 + 15^2} = \frac{5\sqrt{82}}{2}.\end{aligned}$$

APPLICATION - VOLUME OF A PARALLELEPIPED

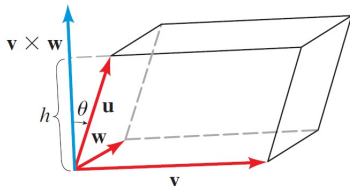
The volume of a parallelepiped determined by $\vec{u}$, $\vec{v}$ and $\vec{w}$ is

$$V = \text{base area} \times \text{height}$$

$$= (\text{area of the parallelogram determined by } \vec{v} \text{ and } \vec{w}) \times ||\vec{u}|| \cos \theta$$

$$= ||\vec{v} \times \vec{w}|| ||\vec{u}|| \cos \theta$$

$$= |(\vec{v} \times \vec{w}) \cdot \vec{u}|.$$



VOLUME OF A PARALLELEPIPED

The volume of a parallelepiped determined by $\vec{u}$, $\vec{v}$ and $\vec{w}$ is

$|(\vec{v} \times \vec{w}) \cdot \vec{u}|$. In particular, if the volume of the parallelepiped is 0, then the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ are **coplanar**, that is, they lie in the same plane.

VOLUME OF A PARALLELEPIPED - EXAMPLE

E.g.: We want to know whether $\vec{u} = \vec{e}_1 + 4\vec{e}_2 - 7\vec{e}_3$,
 $\vec{v} = 2\vec{e}_1 - \vec{e}_2 + 4\vec{e}_3$ and $\vec{w} = -9\vec{e}_2 + 18\vec{e}_3$ are coplanar, so we compute

$$\begin{aligned}(\vec{v} \times \vec{w}) \cdot \vec{u} &= \left(\begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \times \begin{bmatrix} 0 \\ -9 \\ 18 \end{bmatrix} \right) \cdot \begin{bmatrix} 1 \\ 4 \\ -7 \end{bmatrix} \\&= \begin{bmatrix} (-1)(18) - (4)(-9) \\ (4)(0) - (2)(18) \\ (2)(-9) - (-1)(0) \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 4 \\ -7 \end{bmatrix} = \begin{bmatrix} 18 \\ -36 \\ -18 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 4 \\ -7 \end{bmatrix} \\&= (18)(1) + (-36)(4) + (-18)(-7) = 0.\end{aligned}$$

Hence, the three vectors are coplanar.

ACTIVITY 4 [DO IT AT HOME]

(a) Find the area of the $\triangle PQR$.

(i) $P(6, -2, 1)$ $Q(5, -3, -1)$, $R(7, 0, 0)$.

(ii) $P(3, 4, 5)$ $Q(1, 2, 3)$, $R(1, -1, 1)$.

(These are the same points from (b) of activity 3)

(b) Three vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ are given. 1. Find their scalar triple product $\vec{u} \cdot (\vec{v} \times \vec{w})$. 2. Are the vectors coplanar? If not, find the volume of the parallelepiped that they determine.

(i) $\vec{u} = \vec{e}_1 + 2\vec{e}_2 + 3\vec{e}_3$, $\vec{v} = -3\vec{e}_1 + 2\vec{e}_2 + \vec{e}_3$, $\vec{w} = 3\vec{e}_2 + \vec{e}_3$.

(ii) $\vec{u} = \hat{i} - \hat{j} + \hat{k}$, $\vec{v} = -\hat{j} + \hat{k}$, $\vec{w} = 2\hat{i} + \hat{j} - \hat{k}$.

ACTIVITY 4: SOLUTION

(a) (i)

$$\begin{aligned}\text{area of } \triangle PQR &= \frac{1}{2} \|\vec{PQ} \times \vec{PR}\| = \frac{1}{2} \left\| \begin{bmatrix} 5 \\ -3 \\ -1 \end{bmatrix} \right\| \\ &= \frac{\sqrt{5^2 + (-3)^2 + (-1)^2}}{2} = \frac{\sqrt{35}}{2}.\end{aligned}$$

(ii)

$$\begin{aligned}\text{area of } \triangle PQR &= \frac{1}{2} \|\vec{PQ} \times \vec{PR}\| = \frac{1}{2} \left\| \begin{bmatrix} -2 \\ -4 \\ 6 \end{bmatrix} \right\| \\ &= \frac{\sqrt{(-2)^2 + (-4)^2 + 6^2}}{2} = \frac{\sqrt{56}}{2} = \sqrt{14}.\end{aligned}$$

ACTIVITY 4: SOLUTION

(b) (i)

$$\begin{aligned}\vec{u} \cdot (\vec{v} \times \vec{w}) &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \left(\begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} (2)(1) - (1)(3) \\ (1)(0) - (-3)(1) \\ (-3)(3) - (2)(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 3 \\ -9 \end{bmatrix} \\ &= (1)(-1) + (2)(3) + (3)(-9) = -22.\end{aligned}$$

Since $\vec{u} \cdot (\vec{v} \times \vec{w}) \neq 0$, the three vectors are not coplanar.
The volume of the parallelepiped determined by them is
 $|\vec{u} \cdot (\vec{v} \times \vec{w})| = 22$.

ACTIVITY 4: SOLUTION

(b) (ii)

$$\begin{aligned}\vec{u} \cdot (\vec{v} \times \vec{w}) &= \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \cdot \left(\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} (-1)(-1) - (1)(1) \\ (1)(2) - (0)(-1) \\ (0)(1) - (-1)(2) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} \\ &= (1)(0) + (-1)(2) + (1)(2) = 0.\end{aligned}$$

Since $\vec{u} \cdot (\vec{v} \times \vec{w}) = 0$, the three vectors are coplanar.

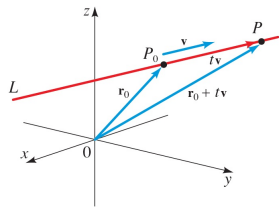
OUTLINE

- ① Three-Dimensional Coordinate Geometry
- ② Vectors in Three Dimensions
- ③ The Cross Product
- ④ Equations of Lines and Planes
- ⑤ Summary

EQUATIONS OF LINES

A line L in 3D space is determined when we know a point $P_0(x_0, y_0, z_0)$ on L and the direction of L . In 3D the direction of a line is described by a vector $\vec{v} = a\vec{e}_1 + b\vec{e}_2 + c\vec{e}_3$ parallel to L . If we let $\vec{r}_0$ be the position vector of P_0 (that is, the vector $\overrightarrow{OP_0}$), then for all real numbers t the terminal points $P(x, y, z)$ of the position vectors $\vec{r}_0 + t\vec{v}$ trace out a line parallel to $\vec{v}$ and passing through P_0 . Each value of the parameter t gives a point P on L . So the line L is given by the position vector $\vec{r}$, where

$$\vec{r} = \vec{r}_0 + t\vec{v} \quad \text{for all } t \in \mathbb{R}$$
$$\Leftrightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad \text{for all } t \in \mathbb{R}.$$

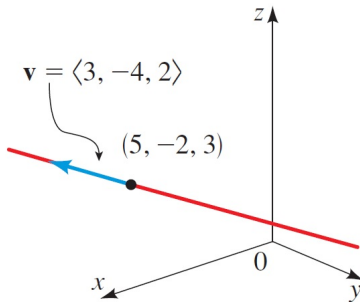


This is the **vector equation of a line**.

EQUATIONS OF LINES - EXAMPLE

E.g.: Given that line L passes through the point $(5, -2, 3)$ and is parallel to $\vec{v} = 3\vec{e}_1 - 4\vec{e}_2 + 2\vec{e}_3$, then its vector equation is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix} + t \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix} \quad \text{for all } t \in \mathbb{R}.$$



EQUATIONS OF PLANES

A plane Π in 3D is determined by a point $P_0(x_0, y_0, z_0)$ in Π and a vector $\vec{n} = a\vec{e}_1 + b\vec{e}_2 + c\vec{e}_3$ that is orthogonal to Π , called a **normal vector** to Π , because for any point $P(x, y, z)$ on Π , the vector $\overrightarrow{P_0P}$ is orthogonal to $\vec{n}$. If we denote the position vectors of P and P_0 by $\vec{r}$ and $\vec{r}_0$ respectively, then it's equivalent to

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$$

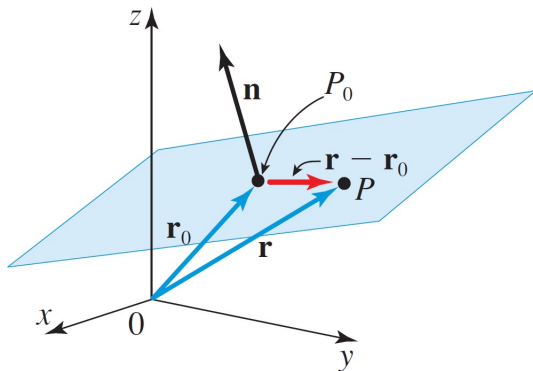
$$\Leftrightarrow \vec{n} \cdot \overrightarrow{P_0P} = 0$$

$$\Leftrightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} = 0$$

$$\Leftrightarrow a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$\Leftrightarrow ax + by + cz = ax_0 + by_0 + cz_0.$$

EQUATIONS OF PLANES



$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$ is called the **vector equation of the plane**.

$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ is called the **standard equation of the plane**.

EQUATIONS OF PLANES - EXAMPLE

E.g.: A plane has normal vector $\vec{n} = 4\vec{e}_1 - 6\vec{e}_2 + 3\vec{e}_3$ and passes through $P(3, -1, -2)$. Then the vector equation of the plane is

$$\begin{bmatrix} 4 \\ -6 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} x - 3 \\ y - (-1) \\ z - (-2) \end{bmatrix} = 0 \Leftrightarrow \begin{bmatrix} 4 \\ -6 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} x - 3 \\ y + 1 \\ z + 2 \end{bmatrix} = 0.$$

The standard equation of the plane is

$$4(x - 3) - 6(y + 1) + 3(z + 2) = 0.$$

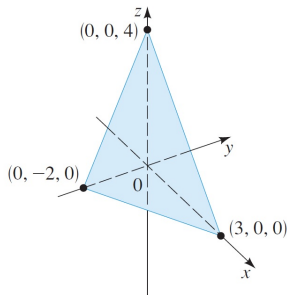
The x -intercept of the plane is at the point in the plane with $y = z = 0$, so $4(x - 3) - 6(0 + 1) + 3(0 + 2) = 0 \Rightarrow x = 3$.

The y -intercept of the plane is at the point in the plane with $x = z = 0$, so $4(0 - 3) - 6(y + 1) + 3(0 + 2) = 0 \Rightarrow y = -2$.

The z -intercept of the plane is at the point in the plane with $x = y = 0$, so $4(0 - 3) - 6(0 + 1) + 3(z + 2) = 0 \Rightarrow z = 4$.

EQUATIONS OF PLANES - EXAMPLE

With the three intercepts, we can sketch the graph of the plane
 $4(x - 3) - 6(y + 1) + 3(z + 2) = 0$.



Note that the plane extends infinitely in all directions without boundaries. We draw it as a triangle to illustrate the intercepts and the intersections of the plane with the xy , yz and xz -planes. Just like when we sketch the graph of a function in the Cartesian plane, we have to stop somewhere as it's impossible to sketch the entire graph over an infinite interval domain.

GENERAL EQUATIONS OF PLANES

We can simplify the standard equation of a plane

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

to

$$ax + by + cz = ax_0 + by_0 + cz_0.$$

Note that the right-hand-side $ax_0 + by_0 + cz_0$ is just a constant. If we denote it as d , then we get the **general equation of the plane**:

$$ax + by + cz = d.$$

E.g.: From the previous example, the general equation of the plane $4(x - 3) - 6(y + 1) + 3(z + 2) = 0$ is

$$4x - 6y + 3z = 12.$$

GENERAL EQUATIONS OF PLANES

Given a general equation $ax + by + cz = d$ of a plane Π , it's very easy to find a point $P_0(x_0, y_0, z_0)$ in Π and a normal vector $\vec{n}$ to Π .

For the former, simply substitute any two of the three variables in $ax + by + cz = d$ by explicit values, say $x = x_0$, $y = y_0$, and then solve for the remaining variable, say solve for z as z_0 .

For the latter, we can simply choose $\vec{n} = a\vec{e}_1 + b\vec{e}_2 + c\vec{e}_3$.

E.g.: Given the general equation of a plane Π is

$$8x + 3y - 3z = 2,$$

then $\vec{n} = 8\vec{e}_1 + 3\vec{e}_2 - 3\vec{e}_3$ is a normal vector to Π . If we let $y = z = 0$, we get $8x + 3(0) - 3(0) = 2 \Leftrightarrow x = \frac{1}{4}$. Thus, the point $P\left(\frac{1}{4}, 0, 0\right)$ lies on Π .

ACTIVITY 5

- (a) Find the vector equation of the line that passes through the points $P(-1, 2, 6)$ and $Q(1, -3, 2)$.
- (b) Find the general equation of the plane that has normal vector $\vec{n} = 3\vec{e}_1 + 2\vec{e}_2 - 1\vec{e}_3$ and passes through $P(0, 2, -3)$.
- (c) Find the general equation of the plane that passes through $P(6, -2, 1), Q(5, -3, -1), R(7, 0, 0)$.
Hint: (b)(i) of activity 3.
- (d) Given two planes

$$\Pi_1 : x - y + 4z = 6 \quad \text{and} \quad \Pi_2 : 7x + 3y - 5z = 1,$$

what is the angle between them?

Hint: the angle between two planes is the acute angle between their normal vectors.

ACTIVITY 5: SOLUTION

(a) The vector equation of the line is

$$\begin{aligned}\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \overrightarrow{OP} + t\overrightarrow{PQ} \\ &= \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} + t \begin{bmatrix} 1 - (-1) \\ -3 - 2 \\ 2 - 6 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 2 \\ 6 \end{bmatrix} + t \begin{bmatrix} 2 \\ -5 \\ -4 \end{bmatrix} \text{ for all } t \in \mathbb{R}.\end{aligned}$$

(b) The general equation of the plane is

$$3x + 2y - z = 3(0) + 2(2) - (-3) \Leftrightarrow 3x + 2y - z = 7.$$

ACTIVITY 5: SOLUTION

(c) From (b)(i) of activity 3, a normal vector to the plane is

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{bmatrix} 5 \\ -3 \\ -1 \end{bmatrix}, \text{ so the general equation of the plane}$$

is

$$5x - 3y - z = 5(7) - 3(0) - (0) \Leftrightarrow 5x - 3y - z = 35.$$

(d) A normal for Π_1 is $\vec{n}_1 = \vec{e}_1 - \vec{e}_2 + 4\vec{e}_3$ and a normal for Π_2 is $\vec{n}_2 = 7\vec{e}_1 + 3\vec{e}_2 - 5\vec{e}_3$. So the angle between the two planes is

$$\begin{aligned} \theta &= \arccos \left(\frac{|\vec{n}_1 \cdot \vec{n}_2|}{\|\vec{n}_1\| \|\vec{n}_2\|} \right) \\ &= \arccos \left(\frac{|(1)(7) + (-1)(3) + (4)(-5)|}{\sqrt{1^2 + (-1)^2 + 4^2} \sqrt{7^2 + 3^2 + (-5)^2}} \right) \\ &= \arccos \left(\frac{|-16|}{\sqrt{1494}} \right) \approx 1.144 \text{ rad} \approx 65.55^\circ. \end{aligned}$$

OUTLINE

- ① Three-Dimensional Coordinate Geometry
- ② Vectors in Three Dimensions
- ③ The Cross Product
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- ⑤ Summary

SUMMARY

- Three-Dimensional Coordinate Geometry
 - the 3D rectangular coordinate system.
 - surfaces in 3D space.
 - distance formula in 3D.
 - the equation of a sphere in 3D.
- Vectors in Three Dimensions
 - vectors in 3D, magnitude/length.
 - algebraic operations on vectors in 3D, unit vector.
 - the dot product for vectors in 3D, the dot product theorem and orthogonal vectors in 3D.
 - the component and projection of vectors in 3D.
- The Cross Product
 - the cross product for vectors in 3D.
 - properties of the cross product.
 - geometry of the cross product.
 - applications of the cross product - area of a parallelogram and volume of a parallelepiped.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 9.3, 9.4, 9.5, 9.6.